INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

CONTAINER ASSEMBLIES, INFLATABLE LIFE RAFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for inflatable life raft and accessory container assemblies.

1.2 Classification. Inflatable life raft containers will be of the following types.

Type I	-	Container Assembly, Life Raft, Helicopter
Type II	-	Container Assembly, Outer, Standard Soft Pack
Type III	-	Container Assembly, Combination Life Raft and Equipment Container
Type IV	-	Container Assembly, Rigid Seat Survival Kit or High Speed Soft Pack
Type V	-	Container Assembly, Pararaft Kit for NB-7 Parachute
Type VI	-	Accessory Container Assembly, LRU-12/A, LRU-13/A, and LRU-1/P
Type VII	-	Accessory Container Assembly, LRU-14/A
Type VIII	-	Accessory Container Assembly, LRU-15/A (Mark 20)
Type IX	-	Accessory Container Assembly, 20 Man Life Raft, for Wing Installation in KC-130F and C-130F Aircraft
Type X	-	Container assembly, VEE Bottom Life Raft Helicopter

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Philadelphia, ATTN: DSCP-NASA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 or email to dscpg&ispecomments@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3 and section 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

MILITARY STANDARDS

MIL-STD-130 - Identification Marking of U.S. Military Property

MIL-STD-889 - Dissimilar Metals

(Copies of these documents are available online at https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAIR DRAWINGS

60A116E4	- Container Assembly, Outer HSSP (High Speed Soft Pack)
62A82D7	- Accessory Container Assembly, LRU-12/A, 13/A & 1/P Life Rafts
62A96D8	- Accessory Container Assembly, LRU-14A/A Life Raft
63A80H4	- Accessory Container Assembly, LRU-15/A (Mark 20) Life Raft
63A80H11	- Accessory Container Assembly, 20 Man Life Raft, for Wing Installation in KC-130F and C-130F Aircraft
68A77H1	- Container Assembly, Life Raft, Helicopter
68A77D2	- Container Assembly, Outer, Standard Soft Pack
68A77H3	- Container Assembly, Combination Life Raft and Equipment Container
68A77D4	- Container Assembly, Rigid Seat Survival Kit or High Speed Soft Pack
1332AS111	- Container Assembly, VEE Bottom Life Raft Helicopter

(Copies of these drawings are available from Naval Air Systems Command, 47123 Buse RD, STE IPT, Patuxent River, MD 20670-1547.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

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(Copies of this document are available from www.asq.org or the American Society for Quality, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)

ASTM INTERNATIONAL

ASTM D6193 - Standard Practice for Stitches and Seams

(Copies of this document are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained

3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the inflatable life raft container assemblies furnished under this specification shall be products which have been inspected and have passed the first article inspection specified in 4.2.

3.2 Material and components. The materials and components shall conform to the applicable specifications, standards, and drawings as listed or required herein (see 6.2). Materials and components which are not covered by applicable specifications, standards, or drawings, or are not specifically described herein, shall be of the lightest practicable weight and entirely suitable for the purpose intended. Unless otherwise specified, the materials and components, except for the metallic parts used in the fabrication of the carrying cases, shall have been manufactured not more than 18 months prior to the date of delivery of the carrying case (see 6.2).

3.2.1 Metal parts. All metal parts shall be of a corrosion resistant material or treated in a manner to render them adequately resistant to corrosion.

3.2.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3 Design and construction. The design and construction of the container assembly shall conform to drawing 60A116E4, 68A77H1, 68A77D2, 68A77H3, 68A77D4, 62A82D7, 62A96D8, 63A80H4, 63A80H11, or 1332AS111, as applicable (see 6.2).

3.3.1 Cut edges. All the cut edges of the uncoated nylon fibrous materials, except for the base cloths which are to be coated, shall be seared, prior to the fabrication of the containers to prevent fraying. No sharp edges shall be formed by the searing process.

3.3.2 Use of the adhesive. In all the cementing operations, the surface to which the adhesive is applied shall be thoroughly cleaned with a suitable solvent in such a manner that the dusting material (talc) and any other surface contaminant is removed. The surface shall be clean prior to cementing. Care should be exercised to insure that the coating and the base undercloth are not damaged, and the adhesion between both is not impaired in any way, or by prolonged ex-posure to the solvent. The solvent used shall evaporate completely prior to the application of the adhesive, and shall leave no residue. Cemented areas shall not contain trapped air, channels, or excessive wrinkling. The adhesive shall be controlled to insure that old adhesive, or adhesive that has partly or completely polymerized is not used. A fresh batch of the adhesive shall be used at least every eight hours during the course of manufacturing (see 4.3.1.1.2). Containers for the adhesive shall be free from congealed adhesive before being refilled.

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3.3.3 Cementing of the seams, seam tapes, patches, and attachments. Seam construction and seam tape, patch, and attachment applications shall be in accordance with applicable drawing requirements. The techniques and precautions outlined in 3.3.2 shall be used in order to effect optimum bonding properties, and to ensure compliance with the requirements of this specification. The adhesive shall be applied in a straight line parallel to the edges of the seam, seam tape, patch, or attachment; and shall extend from just being visible to a maximum of one inch beyond the applicable edge. The adhesive shall not be allowed to remain in clots or permitted to extend in such a manner as to result in localized stiffness. There shall be no spots or stains resulting from the cementing operation. The cemented areas shall not contain any trapped air, channels, edge lifts, or excessive wrinkling.

3.3.4 Seams and stitching. Unless otherwise specified, all the seams and stitching used in the fabrication of the container assemblies shall conform to applicable drawing requirements; and shall be accomplished with stitch Type 301 conforming to ASTM D6193, with 6 to 8 stitches per inch. All the seam edges shall be properly forced out and shall not contain any folds. No seam shall be twisted, puckered, or pleated, and no part of the container shall be caught in an unrelated operation or seam. All the thread ends shall be trimmed to a length of 1/4 to 3/8 inch. Each row of stitching shall be straight and parallel to the seam edge. The straightness of the stitching, in any row, shall be maintained within a tolerance of 1/32 inch. The ends of the stitching that are not caught in other seams or stitching shall be securely backstitched, by overlapping on itself for at least 1/2 inch. Thread breaks, skips, and run-offs shall be overstitched not less than one inch. The thread tension shall be maintained so that there shall not be any loose or overly tight stitching, and the lock shall be imbedded in the materials sewn together. There shall be no spots or stains resulting from the sewing operation.

3.3.4.1 Bartacking. Unless otherwise specified, the number of stitches per bartack shall be based proportionally on 7/8 inch long bartack containing a minimum of 49 stitches. The length and location of each bartack shall be as specified in applicable drawings.

3.4 Color. The color of the containers and components shall be as specified in applicable drawings.

3.5 Markings. Markings shall be in accordance with MIL-STD-130. Ink, color, size, and location of markings shall be as specified in applicable drawings. Letters and numerals 1/4 inch or less in height shall not be stenciled. Serial number for containers shall be in accordance with 3.5.1. All markings shall be thoroughly dry prior to packing or packaging. Markings may be applied directly to the specified area or to a patch securely adhered to the specified area. Material and color of the patch, if used, shall be the same as the container material to which applied.

3.5.1 Serial numbers. Unless otherwise specified, each container shall be identified by a serial number which shall be assigned by the manufacturer. Serialization shall be by a block of consecutive number to cover the entire acquisition quantity.

3.6 Dimensions. When measured as specified in 4.4.2, the dimensions of the containers shall conform to applicable drawing requirements.

3.7 Workmanship. After completion of the final assembly, the containers shall be thoroughly cleaned and all loose thread, lint, and foreign matter shall be removed. The grommets and snap fasteners shall be clinched without distortion, damage, splitting, or cutting of the cloth, webbing or tape as applicable. The metal components shall not be misaligned nor contain any sharp edge, crack, dent, nick, burr, or sliver. The containers shall not contain any nonspecified hole, abraded area, tear, cut, mend, needle chew, spot, or stain. The container assemblies shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

4. VERIFICATION

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4.1 Classification of inspection. The examination and testing of the life raft container assemblies shall be classified as follows:

(a) First article inspection. First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirement of this specification (see 3.1, 4.2, 4.2.1, and 4.2.1.1).

(b) Conformance inspection. Conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification (see 4.3 through 4.3.1.2).

4.2 First article inspection. The first article inspection of the inflatable life raft container assemblies shall consist of all the examinations and tests of this specification.

4.2.1 First article samples. Unless otherwise specified, as soon as practicable after the award of the contract or order, the manufacturer shall submit two container assemblies of the type specified in the contract or order. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production of the type container specified, from contract to contract, submission of additional first article samples on the new contract may be waived at the discretion of the procuring activity (see 6.2). Approval of the first article inspection samples or the waiving of the first article inspection does not preclude the requirements for performing the conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2).

4.2.1.1 First article sample disposition. Upon completion of the first article inspection, recommendations and comments pertinent for use in monitoring production will be forwarded by the Government activity responsible for the inspection program (see 6.2) to the contracting officer. One of the approved first article samples will be returned to the manufacturer for reference in monitoring production. The other sample will be consumed or destroyed in the inspection program, and shall not be considered as a part of the quantity to be delivered under the contract.

4.3 Conformance inspection. The sampling and inspection levels shall conform to ASQ Z1.4. The conformance inspection shall consist of the following:

Visual examination
Dimensions

4.3.1 Sampling.

4.3.1.1 Inspection lot.

4.3.1.1.1 Container assembly. An inspection lot size shall be expressed in units of one container assembly of one type made essentially under the same conditions and from the same materials and components. The sample unit shall be one container assembly of one specific type.

4.3.1.1.2 Adhesive. An inspection lot size shall be expressed in units of one batch of adhesive. An inspection lot shall consist of all the batches of the adhesive used by the manufacturer during one day's production of the container assemblies (see 3.3.2). The sample unit shall be one batch of the adhesive used during one day's production of the container assemblies.

4.3.1.2 Sampling for examinations and tests of the container assemblies. The sample size, examinations, and tests required for the container assemblies shall be specified in Table I. However, the Acceptance Quality Limit (AQL) shall be as specified in the contract.

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TABLE I. Sample size, examinations, and tests of the container assemblies.

Inspection	Paragraph		Sample size
	Requirement	Method	
Visual examination	3.2 through 3.5.1, and 3.7, and applicable drawings	4.4.1.1	A. Every container for major defects B. Inspection level II for minor defects 1/
Dimensional check	3.6 and applicable drawings	4.4.1.1 and 4.4.2	Inspection Level S-3 1/

1/ The sample size shall be based on the applicable sample size code letter corresponding to the specified inspection level of ASQ Z1.4.

4.4 Inspection methods.

4.4.1 Visual examination.

4.4.1.1 Container assemblies. Each container assembly of one type, selected as a sample unit from the lot, shall be examined visually to determine conformance to this specification. The classification and list of defects, Table II shall be used to classify and enumerate the defects found.

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TABLE II. Classification of defects for the visual examination of the containers

Defects	Major	Minor
General		
a. Any weakening defect such as smash, multiple float, loose slub needle chew or abraded area:		
(1) Seriously affecting serviceability or appearance	X	
(2) Not seriously affecting serviceability or appearance		X
b. Color of any component not as specified or contains one or more shade bars		X
c. Any thread end not trimmed to a length of 1/4 to 3/8 inch or loose threads ends		X
d. Any spot or stain:		
(1) Seriously affecting serviceability or appearance	X	
(2) Not seriously affecting serviceability or appearance		X
e. Any edge of the uncoated nylon fibrous materials not seared or contains sharp edges	X	
f. Any non-specified hole, cut, tear, mend or burn	X	
Metallic components		
a. Any surface rough, misaligned, or contains any crack, nick, burr, sharp edge, dent, or sliver	X	
b. Any surface unclean or contains embedded foreign matter		X
c. Any loose, detached, or otherwise not securely retained	X	
d. Any malformed, corroded, fractured, broken, chipped, bent, or distorted	X	

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TABLE II (Continued)

Defects	Major	Minor
e. Any improperly finished:		
(1) Affecting serviceability or appearance	X	
(2) Not affecting serviceability or appearance		X
f. Any snap fastener or grommet misplaced or improperly clinched resulting in cutting of the cloth, tape, or webbing, as applicable	X	
g. Any snap fastener mismated or failing to function properly when snapped closed and unsnapping each of the snap fastener at least three times	X	
h. Any part of the slide fastener assembly bent, broken, or the movement of the slider interfered with when checked for proper function by opening and closing the slide fastener at least three times, along its entire length	X	
Components and assembly		
a. Any component not as specified or any defect of a component or defect of assembly not herein classified:		
(1) Affecting serviceability or appearance	X	
(2) Not affecting serviceability or appearance		X
b. Any component, component part, or required operation omitted or any operation improperly performed not herein classified:		
(1) Affecting serviceability or appearance	X	
(2) Not affecting serviceability or appearance		X

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TABLE II (Continued)

Defects	Major	Minor
Seams and stitching		
a. Accuracy of seaming:		
(1) Any seam, hem, or attachment of any component twisted, puckered, pleated, or caught in an unrelated operation or stitching; any seam edge not properly forced out or contains any fold		X
(2) Ends of stitching that are not caught in other seams or stitching:		
(a) Not securely backstitched	X	
(b) Secure but backstitched for a distance less than 1/2 inch or the backstitching is not overlapped on itself		X
(3) Thread breaks:		
(a) Not securely overstitched	X	
(b) Secure but overstitched for a distance less than one inch		X
b. Gage of stitching:		
(1) Any stitching irregular, unevenly gaged, or various stitching not uniform, when observed on major portion of the seam		X
c. Open seams 1/:		
(1) 1/2 inch or less		X
(2) More than 1/2 inch	X	
d. Raw edges 2/:		
(1) More than 1/4 inch but not more than one inch		X
(2) More than one inch	X	
e. Seam and stitch type:		
(1) Not specified seam or stitch type	X	

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TABLE II (Continued)

Defects	Major	Minor
f. Stitch tension 3/:		
(1) Loose tension, resulting in a loose seam or tight tension resulting in breaking of stitches, when normal pull is applied	X	
g. Stitches per inch (only when condition exists on the major portion of the seam):		
(1) Stitches more or less than specified:		
(a) One or two stitches		X
(b) More than two stitches	X	
Cementing operations		
a. Any cementing operation not in accordance with requirements:		
(1) Seriously affecting serviceability or appearance	X	
(2) Not seriously affecting serviceability or appearance.		X
b. Any void or channel in any cementing operation:		
(1) Seriously affecting serviceability or appearance	X	
(2) Not seriously affecting serviceability or appearance		X
c. Excessive wrinkling in any cementing operation		X
Markings		
a. Missing, incomplete, incorrect, illegible, or improperly located		X
b. Ink not adhere to surface or easily flakes off	X	

1/ A seam shall be classified as open, when one or more stitches joining a seam are broken or when two or more continuous skipped stitches or run-offs occur. On double stitched seams, a seam is considered open, when either one or both sides of the seam is open.

2/ Any edge that should be turned under and is not, but is securely caught in the stitching, shall be classified as a raw edge. Any raw edge not securely caught in the stitching shall be classified as an open seam.

3/ Puckering is evidence of tight tension or gathering of the material. When puckering is evident, other than the gathering of the material, the seam or hem, as applicable shall be tested by exerting normal pull in the lengthwise direction of the seam, hem, or stitching. Gathering of the material shall be recorded as a puckered seam or hem, as applicable.

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4.4.2 Dimensions. The container assembly to be measured shall be laid flat, without tension, on a smooth flat surface so that the creases and wrinkles shall not affect the measurement. The dimensions of the container assembly, and components, location of the components, width of the seams and hems, and turn-unders, as applicable, shall be determined by a yardstick, metal tape, or any suitable measuring device.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The containers covered by this specification are intended to be used for storage and protection of one man, multi-place, and vee bottom life rafts and required accessories which are used by air crewman in a sea survival situation.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification, including any amendments.

(b) Type, Government part number, National Stock Number and quantity desired.

(c) Applicable drawings, including revisions (see 3.3).

(d) Whether first article inspection is waived (see 4.2.1).

(e) Name and address of the first article inspection laboratory (see 4.2.1, and the name of the Government Activity responsible for conducting the inspection program (see 4.2.1.1)).

(f) Packaging requirements (see 5.1)

(h) Certificate of compliance for:

(1) Material and components conforming to applicable specifications, standards, and drawings (see 3.2).

(2) The age of the materials and components (see 3.2).

6.3 Data. For the information of contractors and contracting officers, any of the data specified in applicable documents listed in Section 2 of this specification, or referenced lower-tier documents need not be prepared for the Government and will not be furnished to the Government unless specified in the contract or order. The data to be furnished will be listed on DD Form 1423 (Contractor Data Requirements List), which will be attached to and made a part of the contract or order.

6.4 When a first article is required (see 6.2), it should be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.5 Subject term (key word) listing.

Seams
Stowage
Survival equipment

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - AS

Preparing activity:
DLA - IS

(Project 4220-2010-001)

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